

Archive Feature Decisions

Papyrus Document Intelligence System

Date: April 12, 2026

1. Context

Papyrus ingests scanned page images from physical hardbound committee meeting minute books. The archive feature saves the original image and extracted OCR text to the filesystem. Filenames are derived from the document content so each archived file traces back to its physical source.

2. Scope

Archive only OCR-processed files:

- Image files: PNG, JPG, TIFF, BMP, GIF (via ImageOcrExtractor)
- Scanned PDFs: PDFs that fall back to OCR (via SmartPdfExtractor)

Non-OCR documents (digital PDFs, DOCX, XLSX, etc.) are NOT archived.

3. Decisions

Decision	Choice
Archive scope	OCR-processed files only
Storage path	/data/papyrus/archive
Directory layout	Side-by-side: {sourceId}/file.png + file.txt
Date in filename	Meeting date from OCR text (not upload date)
Date format	yyyy-MM-dd (ISO, sorts naturally)
Date fallback	Upload date when no date found in content
Page number source	Extracted from OCR text
Page fallback	Omitted from filename if not found
Content slug	First meaningful line, slugified, max 80 chars
Abbreviations	Configurable in application.yml
Failure behavior	Best-effort: log warning, never block ingestion
Default state	Disabled (opt-in via ARCHIVE_ENABLED=true)

4. Filename Format

{date}_{page}_{content-slug}.{ext}

Example: Page 1 of meeting minutes

Original upload: IMG_001.png

```
{sourceId}/
  1985-01-19_p1_wmm-execomm-minutes.png
  1985-01-19_p1_wmm-execomm-minutes.txt
```

Example: Page 2 (continuation)

Original upload: IMG_002.png

```
{sourceId}/
  1985-01-19_p2_wmm-minutes.png
  1985-01-19_p2_wmm-minutes.txt
```

Fallback Chain

Scenario	Result
All found	1985-01-19_p1_wmm-execomm-minutes.png
No page found	1985-01-19_wmm-execomm-minutes.png
No date in content	2026-04-12_p1_wmm-execomm-minutes.png
No text at all	2026-04-12_IMG_001.png (original name)

5. Extraction Patterns

Date Extraction (first match wins)

Document text	Pattern	Result
January 19, 1985	Month DD, YYYY	1985-01-19
19 January 1985	DD Month YYYY	1985-01-19
1985-01-19	YYYY-MM-DD (ISO)	1985-01-19
01/19/1985	MM/DD/YYYY	1985-01-19

Page Number Extraction

Document text	Pattern	Result
1.	Standalone number + period	p1
Page 42	Page N	p42
p. 3	p. N	p3
- 5 -	Dash-surrounded number	p5

6. Abbreviation Configuration

Abbreviations are configurable in application.yml and applied to the content slug before filename generation. Case-insensitive replacement.

```
papyrus:
  archive:
    enabled: true
    path: /data/papyrus/archive
    abbreviations:
      "executive committee": excomm
      "western mindanao mission": wmm
```

7. Architecture Summary

Module: papyrus-pipeline (shared between API and MCP)

- Key class: ArchiveService
- archive(sourceId, filename, content, extractedText)
 - Extracts date, page, slug from extractedText
 - Applies abbreviation map
 - Writes original file + .txt to {basePath}/{sourceId}/

- Configuration: PapyrusProperties.ArchiveProperties
- enabled (boolean)
 - path (String)

- abbreviations (Map<String, String>)

Integration points:

- DocumentService (papyrus-api) - after OCR extraction
- IngestionOrchestrator (papyrus-mcp) - after OCR extraction

Docker: Shared volume 'papyrus_archive' mounted at /data/papyrus/archive

Env vars: ARCHIVE_ENABLED, ARCHIVE_PATH